Non-Minimality of Minimal Telescopers Explained by Residues

Yisen Wang

Johannes Kepler Universität Linz

KLMM, AMSS, Chinese Academy of Sciences



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Joint work with Shaoshi Chen, Manuel Kauers, Christoph Koutschan, Xiuyun Li and Rong-Hua Wang

Wilf–Zeilberger theory

In the early 1990s, Wilf and Zeilberger developed an algorithmic theory for proving combinatorial and special function identities.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

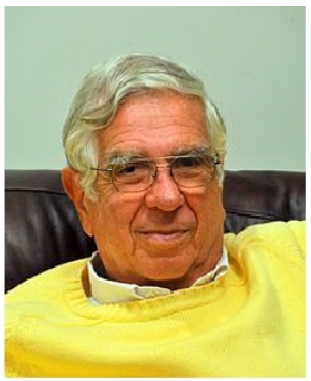
$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$$

$$\sum_{j=0}^{k} \binom{k}{j}^2 \binom{n+2k-j}{2k} = \binom{n+k}{k}^2$$

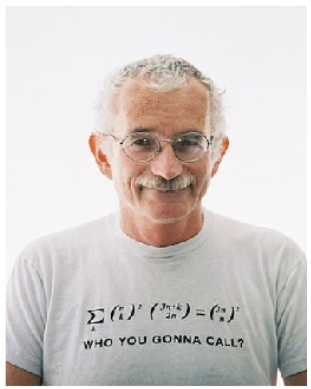
$$\int_{-1}^{+1} \frac{e^{-px} T_n(x)}{\sqrt{1-x^2}} dx = (-1)^n \pi I_n(p)$$

$$\int_0^{\infty} x^{\alpha-1} Li_n(-xy) dx = \frac{\pi(-\alpha)^n y^{-\alpha}}{\sin(\alpha\pi)}$$

$$\ldots$$



Herbert Wilf



Doron Zeilberger

Telescoping

Problem. For a sequence $f(k)$ in some class $\mathfrak{S}(k)$, decide whether there exists $g(k) \in \mathfrak{S}(k)$ (and if so, compute it) such that

$$f(k) = g(k+1) - g(k) =: \Delta_k(g)$$

$$\Downarrow$$

$$\sum_{k=a}^{b} f(k) = g(b+1) - g(a)$$

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Examples.

- Rational sums

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \Delta_k\left(-\frac{1}{k}\right) = 1 - \frac{1}{n+1}$$

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- Hypergeometric sums

$$\sum_{k=0}^{n} \frac{\binom{2k}{k}^2}{(k+1)4^{2k}} = \sum_{k=0}^{n} \Delta_k\left(\frac{4k\binom{2k}{k}^2}{4^{2k}}\right) = \frac{4(n+1)\binom{2n+2}{n+1}^2}{4^{2n+2}}$$

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Hypergeometric:

$$\frac{f(k+1)}{f(k)} = \text{rational.}$$

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Creative telescoping

Problem. For a sequence $f(n,k)$ in some class $\mathfrak{S}(n,k)$, find a linear recurrence operator $L \in C[n, S_n]$ and $g \in \mathfrak{S}(n,k)$ such that

$$\underbrace{L(n, S_n)}_{\text{telescoper}}(f) = \Delta_k(g).$$

where C is a field of char. 0. We call g a corr. certificate for L.

Example. Let $f(n,k) = \binom{n}{k}^2$. Then a telescoper for f and its certificate are

$$L = (n+1)S_n - 4n - 2 \quad \text{and} \quad g = \frac{(2k-3n-3)k^2\binom{n}{k}^2}{(k-n-1)^2}$$

Proving hypergeometric identities

$$F(n) := \sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$$

Creative telescoping for $f = \binom{n}{k}^2$: $L(f) = \Delta_k(g)$, where

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$$L(F(n)) = \sum_{k=0}^{n} L(f(n,k)) + (n+1) \cdot f(n+1, n+1),$$

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In this example, L is the minimal annihilator of $F(n)$.

But this is not always the case!

In general, the minimal telescoper for $f(n,k)$ is not the minimal annihilator of $F(n)$, but just a left multiple.

Non-minimality phenomenon

Example (Vanishing sum).

$$\sum_{k=0}^{2n+1} (-1)^k \binom{2n+1}{k}^2 = 0.$$

The minimal telescoper is $(2n+3)S_n + (8n+8)$.

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Example (Paule and Schorn, 1995).

$$F(n) := \sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{dk}{n} = (-d)^n, \quad \text{where } d \geq 2.$$

The minimal telescoper for $f = (-1)^k \binom{n}{k} \binom{dk}{n}$ is of order $d-1$, while $F(n)$ satisfies

$$\boxed{(S_n + d)} \cdot F(n) = 0.$$

Van Hoeij's submodule approach

Denote

$$A_{n,k} := C(n,k)[S_n, S_k], \quad A_n := C(n)[S_n].$$

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For a given hypergeometric summand $H(n,k)$, define

$$A_n\text{-module:} \quad \Omega := A_{n,k} \cdot H \quad \text{and} \quad M := \Omega / \Delta_k(\Omega).$$

Then L is a telescoper for H iff it annihilates $\overline{H} := H + \Delta_k(\Omega)$.

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- Let $N < M$ and let $R \in A_n$ be such that

$$R(\overline{H} + N) = 0 \in M/N.$$

  Then the minimal such R is a right factor of L.

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  Then the minimal such R is a right factor of L.
- Let

  $$M = M_1 \oplus M_2 \quad \Rightarrow \quad \overline{H} = \overline{H}_1 + \overline{H}_2.$$

  Suppose L_1, L_2 are the minimal annihilators of $\overline{H}_1, \overline{H}_2$. Then

  $$L = \mathrm{lclm}(L_1, L_2)$$

  is the minimal telescoper for H.

Van Hoeij's explanation of non-minimality

Compute the minimal recurrence of

$$F(n) := \sum_{k=0}^{n} H, \quad \text{where } H := \frac{\binom{n}{k}^7}{2n+3k}.$$

Let

$$R := S_n^3 - \left(\frac{54(n+2)(n+1)n(2n+3)}{5(5n+12)(5n+9)(5n+6)(5n+3)} \right)^7,$$

then $R(\overline{H}) \in N < M$, which implies that R is a right factor of L.

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By observation,

$$\phi : k \to n-k \text{ is an } A_n\text{-automorphism of } N.$$

Van Hoeij's explanation of non-minimality

Then $\phi^2 = 1$ yields

$$N = N_1 \oplus N_2 \quad \Rightarrow \quad R(\overline{H}) = \overline{H}_1 + \overline{H}_2,$$

where

$$\overline{H}_1 = \frac{1+\phi}{2}\big(R(\overline{H})\big) \quad \text{and} \quad \overline{H}_2 = \frac{1-\phi}{2}\big(R(\overline{H})\big).$$

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Let $L_1,\ L_2$ be the minimal annihilators of $\overline{H}_1,\ \overline{H}_2 \in N$, then

$\mathrm{lclm}(L_1, L_2) \circ R$ is the minimal telescoper for H.

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Since $\phi : k \to n-k$, we have $\sum_{k=0}^{n} \overline{H}_2 = 0$ and

$L_1 \circ R$ is the minimal recurrence of $F(n)$.

Multiplicative factorization

Multiplicative form.

$$H = \underbrace{S}_{\text{rational part}} \cdot \underbrace{H_0}_{\text{non-rational part}} \qquad \text{with} \quad \frac{S_k(H_0)}{H_0} =: K.$$

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We require $K := u/v$ to be shift-reduced in k, i.e.,

$$\gcd(u(k), v(k+i)) = 1 \quad \text{for all } i \in \mathbb{Z}.$$

Then we call S a shell and K a kernel of H.

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Remark. $K = 1 \iff H$ is rational in k.

Abramov–Petkovšek reduction

Additive Decomposition. Let $H = S \cdot H_0$ with

$$\frac{S_k(H_0)}{H_0} = K = \frac{u}{v}.$$

Then $\exists g \in C(n)(k)$ and $u, v \in C(n)[k]$ s.t.

$$H = \Delta_k(g \cdot H_0) + \left(\frac{a}{b} + \frac{p}{v}\right) \cdot H_0,$$

where b is shift-free, $\deg_k(a) < \deg_k(b)$, and

$$\gcd\left(b, S_k^{-\ell}(u)\right) = \gcd\left(b, S_k^{\ell}(v)\right) = 1 \quad \text{for all } \ell \geq 0.$$

Remark. $H \in \Delta_k(\Omega) \quad \not\Rightarrow \quad a = p = 0.$

Polynomial reduction

Given $K = u/v \in C(n)(k)$ with $K \neq 0$, define

$$\begin{array}{rccc} \phi_K : & C(n)[k] & \longrightarrow & C(n)[k] \\ & p & \longmapsto & u \cdot S_k(p) - v \cdot p. \end{array}$$

Call ϕ_K the polynomial reduction map w.r.t. K.

Then we have

$$C(n)[k] = \mathrm{im}(\phi_K) \oplus \mathbb{W}_K,$$

$$\mathbb{W}_K = \mathrm{Span}_{C(n)} \left\{ k^i \;\middle|\; \forall q \in \mathrm{im}(\phi_K),\ \deg_k(q) \neq i \right\}.$$

Call $\mathbb{W}_K$ the standard complement of $\mathrm{im}(\phi_K)$.

Let $p = p_1 + p_2$ with

$$p_1 = \phi_K(w) \text{ and } p_2 \in \mathbb{W}_K.$$

Modified Abramov–Petkovšek reduction

For a hypergeometric term $H = S \cdot H_0$, where

$$\frac{S_k(H_0)}{H_0} = K = \frac{u}{v}.$$

Step 1. Abramov–Petkovšek's reduction:

$$H = \Delta_k(g \cdot H_0) + \left(\frac{a}{b} + \frac{p}{v}\right) \cdot H_0,$$

Step 2. After polynomial reduction, we obtain:

$$H = \underbrace{\Delta_k\big((g+w) \cdot H_0\big)}_{\text{summable part}} + \underbrace{\left(\frac{a}{b} + \frac{p_2}{v}\right) \cdot H_0}_{\text{residual form}}.$$

Proposition. H is hypergeometric summable w.r.t. k iff $a = p_2 = 0$.

Kernel submodule and prescoper

Theorem. Let

$$M := \left\{ \left(\frac{a}{b} + \frac{p}{v} \right) \cdot H_0 + \Delta_k(\Omega) \;\middle|\; p \in \mathbb{W}_K \right\},$$

Kernel submodule and prescoper

Theorem. Let

$$N := \left\{ \frac{p}{v} \cdot H_0 + \Delta_k(\Omega) \;\middle|\; p \in \mathbb{W}_K \right\}.$$

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Definition. A nonzero operator $R \in A_n$ is called a prescoper if

$$R(H) = \underbrace{\Delta_k(\cdots)}_{\text{summable part}} + \underbrace{\left(\frac{a}{b} + \frac{p}{v}\right) \cdot H_0}_{\text{residual form}},$$

that is, R cancels the part a/b. Then we get

$$R(H) + \Delta_k(\Omega) \in N \quad \text{i.e.,} \quad R(\overline{H}) \in N.$$

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Lemma. The minimal prescoper is a right factor of the telescoper.

Zero-sum submodules

Idea. Find vanishing sums by the residue approach.

Definition. We call $Z < N$ a zero-sum submodule if

$$\sum_k \frac{p}{v} \cdot H_0 = 0, \text{ for each } \frac{p}{v} \cdot H_0 + \Delta_k(\Omega) \in Z.$$

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Let $R, T \in A_n$ s.t. $R(\overline{H}) \in N$ and $T\big(R(\overline{H})\big) \in Z$, then

$$T \circ R\big(F(n)\big) = T \circ R\left(\sum_k H\right) = \sum_k T \circ R(H) = 0.$$

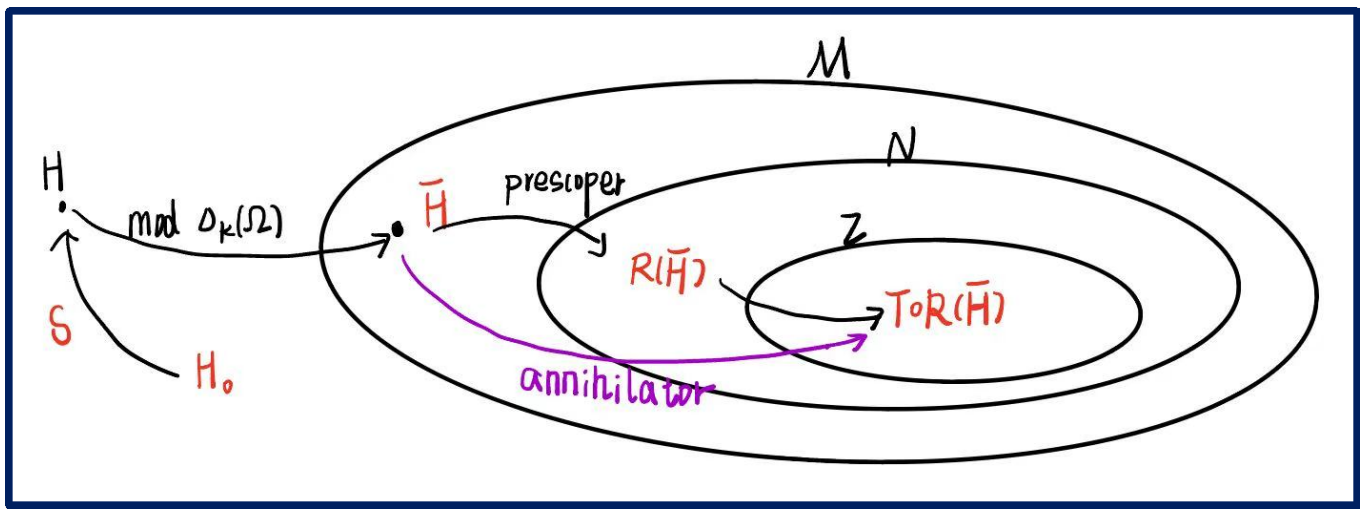
It gives us a way to determine an annihilator for $F(n)$.

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Nicole's theorem

Theorem (Nicole, 1717). Let $P \in \mathbb{C}[x]$ be s.t. $\deg(P) \leq n-2$. Then

$$f = \frac{P(x)}{(x+\beta_1)\cdots(x+\beta_n)}, \quad \text{where } \beta_i - \beta_j \in \mathbb{Z}\setminus\{0\} \text{ for } i \neq j,$$

is summable in $\mathbb{C}(x)$.

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is summable in $\mathbb{C}(x)$.

Proof. By partial fraction decomposition,

$$f = \sum_{i=1}^{n} \frac{\alpha_i}{x+\beta_i}, \quad \text{where } \alpha_i \in \mathbb{C}.$$

Then f is summable iff $\sum_{i=1}^{n} \alpha_i = 0$. By normalizing f, we get

$$P(x) = (\alpha_1 + \cdots + \alpha_n)x^{n-1} + \text{ lower terms.}$$

Since $\deg(P) \leq n-2$, we have $\sum_{i=1}^{n} \alpha_i = 0$.

From Nicole's theorem to vanishing sums

Corollary. Let $P \in \mathbb{C}[x]$ be such that $\deg(P) \leq n-1$. Then

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} P(k) = 0.$$

From Nicole's theorem to vanishing sums

Corollary. Let $P \in \mathbb{C}[x]$ be such that $\deg(P) \leq n-1$. Then

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} P(k) = 0.$$

Proof. Consider the rational function

$$f = \frac{n!P(-x)}{x(x+1)\cdots(x+n)}.$$

By Nicole's theorem, f is summable in $\mathbb{C}(x)$. Then Lagrange's formula implies that

$$\sum_{k=0}^{n} \operatorname{res}_x(f,-k) = \sum_{k=0}^{n} \frac{n!P(k)}{(-1)^k k!(n-k)!} = 0.$$

Example 1

Recall the previous example (Paule and Schorn, 1995):

$$F(n) := \sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{3k}{n} = (-3)^n.$$

The minimal telescoper for $H := (-1)^k \binom{n}{k} \binom{3k}{n}$ is

$$2(2n+3)S_n^2 + 3(5n+7)S_n + 9(n+1).$$

The minimal annihilator for $F(n)$ is $S_n + 3$.

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Decompose H by its shell and kernel,

$$H_0 = H, \quad \frac{S_k(H_0)}{H_0} = \frac{3(k-n)(3k+1)(3k+2)}{(3k-n+1)(3k-n+2)(3k-n+3)} =: \frac{u}{v}.$$

Define the kernel submodule of M,

$$N := \left\{ \frac{p}{v} \cdot H_0 + \Delta_k(\Omega) \;\middle|\; p \in \mathbb{W}_K \right\},$$

where $\mathbb{W}_K$ has a $\mathbb{Q}(n)$-basis $\{1, k^3\}$. By reduction,

$$H + \Delta_k(\Omega) \in N \quad \Rightarrow \quad R := 1.$$

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Consider the rational function

$$f = \frac{p(n,-x)(-3x)(-3x-1)\cdots(-3x-n+4)}{x(x+1)\cdots(x+n)},$$

then by Nicole's theorem,

$$\sum_k \frac{p}{v} \cdot H_0 = 0, \text{ when } \deg_k(p) \le 2$$

$$\Downarrow$$

$$Z := \left\{ \frac{p}{v} \cdot H_0 + \Delta_k(\Omega) \;\middle|\; \deg_k(p) = 0 \right\}.$$

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By reduction,

$$H_0 \equiv \frac{(81n+108)k^3 + \text{ terms free of } k}{(3n+4)\cdot v} \cdot H_0 \mod \Delta_k(\Omega).$$

$$S_n(H_0) \equiv \frac{(-243n-324)k^3 + \text{ terms free of } k}{(3n+4)\cdot v} \cdot H_0 \mod \Delta_k(\Omega).$$

Hence, $T := S_n + 3$ and

$$(S_n+3)(\overline{H}) \in Z$$

$$\Downarrow$$

$$(S_n+3)\cdot F(n) = \sum_k (S_n+3)\cdot H = 0.$$

Example 2

Let

$$F(n) := \sum_k H, \quad \text{where } H := (-1)^k \binom{3n+1}{k} \binom{3n-k}{n}^3.$$

The minimal telescoper for H is of order 2.

While $F(n)$ satisfies

$$F(n) = 1 \quad \Rightarrow \quad (S_n - 1) \cdot F(n) = 0.$$

Example 2

Let

$$F(n) := \sum_k H, \quad \text{where } H := (-1)^k \binom{3n+1}{k} \binom{3n-k}{n}^3.$$

Decompose H by its shell and kernel,

$$H_0 = (k-3n-1)\cdot H, \quad \frac{S_k(H_0)}{H_0} = \frac{(k-2n)^3}{(k+1)(k-3n)^2} =: \frac{u}{v}.$$

Define the kernel submodule of M,

$$N := \left\{ \frac{p}{v}\cdot H_0 + \Delta_k(\Omega) \;\middle|\; p \in \mathbb{W}_K \right\},$$

where $\mathbb{W}_K$ has a $\mathbb{Q}(n)$-basis $\{1, k^3\}$. Then

$$R := S_n - 1 \text{ with } R(\overline{H}) \in N.$$

Example 2

Let

$$F(n) := \sum_k H, \quad \text{where } H := (-1)^k \binom{3n+1}{k} \binom{3n-k}{n}^3.$$

Consider the rational function

$$f = \frac{p(n,-x)(3n+1)!(x+3n-1)^2 \cdots (x+2n+1)^2}{(n!)^3 (x-1)x(x+1)\cdots(x+2n)},$$

then Nicole's theorem,

$$\sum_k \frac{p}{v} \cdot H_0 = 0, \text{ when } \deg_k(p) \le 2$$

$$\Downarrow$$

$$Z := \left\{ \frac{p}{v} \cdot H_0 + \Delta_k(\Omega) \;\middle|\; \deg_k(p) = 0 \right\}.$$

Example 2

Let

$$F(n) := \sum_k H, \quad \text{where } H := (-1)^k \binom{3n+1}{k} \binom{3n-k}{n}^3.$$

By reduction,

$$R(H) = S_n(H) - H \equiv \frac{37n^7 + 96n^6 + 81n^5 + 22n^4}{8(n+1)^3(9n^2+10n+3)} \frac{H_0}{v} \mod \Delta_k(\Omega).$$

Hence, $R(\overline{H}) \in Z$ and $T := 1$. Recall that $R = S_n - 1$, so

$$(S_n - 1)(F(n)) = R\big(F(n)\big) = \sum_k R(H) = 0.$$

Summary

- Non-minimality phenomenon
- Construct the submodule for hypergeometric terms
- Vanishing sums via discrete residues

